



Algorithms Approaching the Threshold for Semi-random Planted Clique

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ABSTRACT

We design new polynomial-time algorithms for recovering planted cliques in the semi-random graph model introduced by Feige and Kilian. The previous best algorithms for this model succeed if the planted clique has size at least $n^{2/3}$ in a graph with n vertices. Our algorithms work for planted-clique sizes approaching $n^{1/2}$ — the information-theoretic threshold in the semi-random model and a conjectured computational threshold even in the easier fully-random model. This result comes close to resolving open questions by Feige and Steinhart.

To generate a graph in the semi-random planted-clique model, we first 1) plant a clique of size k in an n -vertex Erdős–Rényi graph with edge probability $1/2$ and then adversarially add or delete an arbitrary number edges not touching the planted clique and delete any subset of edges going out of the planted clique. For every $\varepsilon > 0$, we give an $n^{O(1/\varepsilon)}$ -time algorithm that recovers a clique of size k in this model whenever $k \geq n^{1/2+\varepsilon}$. In fact, our algorithm computes, with high probability, a list of about n/k cliques of size k that contains the planted clique. Our algorithms also extend to arbitrary edge probabilities p and improve on the previous best guarantee whenever $p \leq 1 - n^{-0.001}$.

Our algorithms rely on a new conceptual connection that translates certificates of upper bounds on biclique numbers in *unbalanced* bipartite Erdős–Rényi random graphs into algorithms for semi-random planted clique. Analogous to the (conjecturally) optimal algorithms for the fully-random model, the previous best guarantees for semi-random planted clique correspond to spectral relaxations of biclique numbers based on eigenvalues of adjacency matrices. We construct an SDP lower bound that shows that the $n^{2/3}$ threshold in prior works is an inherent limitation of these spectral relaxations. We go beyond this limitation by using higher-order sum-of-squares relaxations for biclique numbers.

We also provide some evidence that the information-computation trade-off of our current algorithms may be inherent by proving an average-case lower bound for unbalanced bicliques in the low-degree polynomial model.

CCS CONCEPTS

• **Theory of computation** → **Discrete optimization**; **Semidefinite programming**.

KEYWORDS

planted clique, semi-random, semidefinite programming, sum-of-squares hierarchy

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1 INTRODUCTION

Clique is one of the most intensely studied combinatorial problems in theoretical computer science, both in terms of its worst-case and its average-case complexity. It was among the first graph problems shown to be NP-complete [35]. In fact, it turns out that for every $\varepsilon > 0$, it is NP-hard to find cliques of size n^ε even in graphs that contain cliques of size $n^{1-\varepsilon}$ [26, 36, 54].

The most well-studied average-case counterpart is the *planted clique* problem [32, 41] where the goal is to recover a k -clique added to an Erdős–Rényi random graph $G(n, 1/2)$. Such a clique is uniquely identifiable if $k \gg 2 \log_2 n$. There are polynomial time algorithms based on rounding the second eigenvector of the adjacency matrix [1] as well as basic semidefinite programming relaxations (e.g., the Lovász theta function) [19, 20] to recover the planted clique with high probability whenever $k \geq n^{1/2}$. Closing the exponential gap between the information-theoretic threshold value of k and the threshold of the best known algorithms is a tantalizing open question that has inspired a large body of research, culminating in lower bounds against restricted classes of algorithms, such as statistical query algorithms [22] and sum-of-squares relaxations [9], that vastly generalize the current algorithms for this problem. These concrete lower bounds provide some rigorous evidence that current algorithms for planted clique are optimal.

Fragility of algorithms. Unfortunately, many algorithms for the planted clique problem are *fragile*: a small number of adversarial changes to the input can cause the natural algorithms to break down completely. This includes methods based on basic statistics such as degrees of vertices or eigenvalues of the adjacency matrix that provide the strongest possible guarantees for the problem. Such fragility can be viewed as known algorithms *overfitting* to the choice of the distributional model.

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In response, a significant research effort has gone into finding algorithms resilient against even the most benign forms of adversarial modifications. This includes a long line of work on *monotone adversary* models introduced in [17] for average-case formulations of clique and coloring (i.e., community detection) [44, 46, 47]. In the context of planted clique, such models correspond to starting from the standard planted clique input and allowing an adversary to *delete* any subset of edges not in the planted clique. Such deletions are, in principle, only helpful since the planted clique continues to be the true maximum clique in the resulting graph. And indeed, while basic statistics and spectral methods fail in the presence of monotone adversaries, natural analyses of more resilient algorithms based on semidefinite programming [19] succeed at the same $k = O(\sqrt{n})$ threshold while tolerating *monotone adversaries*.

1.1 Semi-random Model

A seminal work by Feige and Kilian [18] introduced the following semi-random planted clique model following the classical work of [12] on semi-random coloring. Such semi-random models combine a distributional input with a monotone adversary and an adversarial choice at the same time. After the introduction of this model, similar semi-random models have been studied for a wide range of combinatorial optimization problems, including graph partitioning and constraint satisfaction problems. We refer the interested reader to the excellent survey [16].

Definition 1.1 (Feige–Kilian semi-random planted clique model, $\text{FK}(n, k, p)$). For $n, k \in \mathbb{N}$ with $k \leq n$ and $p \in [0, 1]$, we let $\text{FK}(n, k, p)$ be the collection of distributions over graphs with vertex set $V = [n]$ sampleable by a process of the following form:

- (1) *Random Generation Phase*: Choose a uniformly random subset $S^* \subseteq V$ of size k and add a clique on S^* to an Erdős–Rényi random graph $G(n, p)$ (which includes each possible edge independently at random with probability p),
- (2) *Adversarial Deletion Phase*: delete an arbitrary subset of edges going out of S^* adaptively (i.e., possibly depending on the previous random choices),
- (3) *Adversarial Addition Phase*: replace the subgraph induced on $V \setminus S^*$ by an arbitrary one, again adaptively.

Unlike planted clique with monotone adversaries, semi-random models are far from “helpful”. In particular, the planted clique isn’t necessarily the maximum clique in the resulting graph. And the adversarial choices in the generation process are known to result in significantly altered information-theoretic thresholds at which efficient algorithms can succeed for related problems such as community detection in the stochastic block model [46].

If $p = 1$, the above model recovers the worst-case version of the clique problem. On the other hand, by omitting the last two steps, we recover the original planted clique model, and by omitting only the last step we recover the planted clique with “helpful” monotone adversaries. Importantly, the last two steps are *adaptive* and can be chosen adversarially in response to the first step. In absence of adaptivity (i.e., when the last two steps are oblivious to the distributional choices), the model becomes significantly easier algorithmically.

We write $G \sim \text{FK}(n, k, p)$ to denote a graph sampled according to one of the distributions in $\text{FK}(n, k, p)$. For particular choices

of parameters $k = k(n)$ and $p = p(n)$, our goal is to develop an algorithm that succeeds with high probability for *every* distribution described by $\text{FK}(n, k, p)$.

What does it mean for the algorithm to succeed? Since the graph induced on $V \setminus S^*$ could be a worst-case hard instance for the clique problem, it is NP-hard to find a maximum clique in G . So the goal of the algorithm is to find a clique of size k in G . For the original planted clique model (and the version with helpful monotone adversaries), we could with high probability recover the planted clique S^* in G . In contrast, in the semi-random model, this task is impossible information-theoretically because the adversary could simulate multiple disjoint copies of the distributional process in $V \setminus S^*$. Instead, we can ask the algorithm to compute a small list of (pairwise almost-disjoint) k -cliques in G that contains the planted clique S^* . Such a list also allows uniquely identifying S^* if, in addition, we are given a random vertex of S^* as advice.

In their work introducing this model, [18] gave an algorithm that uses a Gaussian rounding [25] of the vector solution for the Lovász theta SDP relaxation combined with a combinatorial cleanup step to produce a correct list. For any p such that $1 - p \geq (1 + \epsilon) \ln(n)/n$, their algorithm works if $k \geq \alpha n$ for some constant $\alpha > 0$. Such a guarantee is essentially optimal if $1 - p = \tilde{O}(1/n)$. The main focus of subsequent works has been in the case when $1 - p$ is larger. In particular, the case of $p = 1/2$ (and more generally, any constant < 1) is of special interest. In this case, one can ideally expect polynomial time algorithms that succeed for $k \sim \sqrt{n}$ as in the case of average-case planted clique. We will focus on the case of $p = 1/2$ in this introduction for the sake of clarity.

Prior work. Algorithms in prior works rely on rounding carefully designed semidefinite programming (SDP) relaxations. In the slightly easier setting that drops the adversarial deletion step from the model, Charikar, Steinhardt and Valiant [13] gave an algorithm based on a semidefinite programming relaxation for *list-decodable mean estimation* that succeeds whenever $k \geq O(n^{2/3} \log^{1/3}(n))$. Their guarantee was improved to $k \geq O(n^{2/3})$ by Mehta, Mackenzie and Trevisan [45]. The algorithm of [45] is based on a variant of the Lovász theta SDP (that they call “crude” or C-SDP) with an objective function that incentivizes “spread-out” vector solutions and analyzed via the Grothendieck inequality. They suggest (though don’t prove) that their SDP should fail if $k = o(n^{2/3})$. Further heightening the intrigue, Steinhardt [52] proved that if $k = o(\sqrt{n})$, then it is information-theoretically impossible to identify an $O(n/k)$ -size list, indicating an *information-theoretic* (as opposed to computational) phase transition at $k \sim \sqrt{n}$.

Feige’s open question. Given the apparent barrier for the basic semidefinite program at $k \sim n^{2/3}$, it is natural to ask: is the semi-random variant harder than the average-case planted clique problem or could there be algorithms that succeed for k approaching the $O(\sqrt{n})$ threshold? In his survey on semi-random models [16], Feige posed (see Section 9.3.4, Page 205) this as an outstanding open question and hoped for algorithms for semi-random planted clique matching the $k \sim \sqrt{n}$ threshold for the average-case variant.

1.2 Results

In this work, we nearly resolve Feige’s question and give an algorithm for the semi-random planted clique problem that works for k approaching $\sqrt{n}$. Specifically, we give a scheme of algorithms that, for any $\varepsilon > 0$, run in time $n^{O(1/\varepsilon)}$ and succeed in solving the semi-random planted clique problem whenever $k \geq n^{1/2+\varepsilon}$.

THEOREM 1.2 (MAIN RESULT). *For every $\varepsilon > 0$, there is an algorithm that, given a graph G on n vertices as input, computes a list of vertex subsets in time $n^{O(1/\varepsilon)}$ satisfying the following guarantee: If G is generated according to $\text{FK}(n, k, 1/2)$ for $k \geq n^{1/2+\varepsilon}$, then with probability at least 0.99 the algorithm outputs a list of at most $(1+o(1))\frac{n}{k}$ cliques of size k such that one of them is the clique planted in G .*

In particular, our algorithm manages to recover planted cliques of size k approaching $\sim \sqrt{n}$ — the information-theoretic threshold [52] and the conjectured computational threshold even for the easier fully-random planted clique problem. This improves on the best known prior algorithm [45] that gives a polynomial time algorithm that succeeds whenever $k \geq O(n^{2/3})$.

Our approach extends to edge probabilities p beyond the choice $p = 1/2$ and yields improved guarantees even when $p = 1 - o(1)$, though in that case we do not approach the information-theoretic threshold. Our hardness results (discussed below) show that such an outcome might be inevitable.

Higher degree sum-of-squares vs basic SDP. Our algorithm relies on rounding a high constant degree sum-of-squares relaxation (that maximizes a natural “entropy-like” objective function) of the natural integer program for finding k -cliques in graphs. As we discuss below, this is likely necessary as for the natural certification problem (discussed below) that arises in algorithms for semirandom planted clique, the basic SDP has a lower bound that precludes recovering $k \ll n^{2/3}$ -size cliques. This is in sharp contrast to the average-case planted clique problem where no constant degree strengthening of the basic SDP allows recovering planted cliques of size $k = o(\sqrt{n})$ [9] (i.e., asymptotically smaller than the threshold for recovery using the basic SDP). In fact, the best known analyses can only obtain an $n^{O(t)}$ algorithm that succeeds whenever $k \geq O(\sqrt{n/2^t})$ [20].

Indeed, few combinatorial optimization problems are known to benefit from high constant degree sum-of-squares relaxations. Some notable exceptions include approximating constraint satisfaction problems on graphs with small threshold rank [2, 10] (where the high degree corresponds to the threshold rank) and approximating the maximum bisection in a graph [48] (where the high degree helps deal with the cardinality constraint). Our work adds a new example to this list that appears to be more unstructured than earlier examples.

We note that, in contrast to combinatorial optimization, in statistical estimation higher degree sum-of-squares relaxations have recently been pivotal in algorithmic applications such as robust method of moments [40], linear regression [7, 37], list-decodable learning [6, 31, 34, 40, 49, 50], and settling the robust clusterability and learnability of high-dimensional Gaussian mixtures [3–5, 29, 40, 43].

Rounding and connection to certifying bicliques. Our rounding algorithm is reminiscent of the “rounding by votes” strategy employed in several recent works on list-decodable learning [6, 31, 34]. Our analysis relies on a new connection to efficient certificates: we can recover a small list which includes the planted clique if we can certify that the planted clique has small intersection with all other k -cliques. This reduces to certifying upper bounds on bipartite cliques in unbalanced bipartite random graphs: see Definition 2.4 for a standalone definition of the problem.

Given a bipartite graph $H = (U, V, E)$ with $|U| = k$, $|V| = n$, and each bipartite edge included in E with probability p independently, it is easy to prove by a standard application of Chernoff and union bounds that there is no ℓ by k bipartite clique in H with $\ell \gg O(\log n / (1 - p))$. The bipartite clique certification problem asks to find a polynomial time verifiable certificate that H contains no ℓ by k biclique for ℓ as small as possible. This is a variant of the more standard biclique certificate problem (see, e.g., [21]) where both the graph and the cliques we are interested in are unbalanced.

Our main result is based on the following primitive that certifies in time $n^{O(1/\varepsilon)}$ that a random k by n bipartite graph does not contain ℓ by k bicliques for $\ell = n^\varepsilon$ and any $k \geq \tilde{O}(\sqrt{n})$. Our certificates are based on $O(1/\varepsilon)$ -degree sum-of-squares proofs and this is necessary — we prove that for any $\ell = o(n/k)$, there are no degree 2 sum-of-squares (i.e., basic SDP) certificates of absence of bicliques. In particular, unlike the case of balanced bipartite random graphs, the unbalanced setting seems to naturally benefit from large constant degree sum-of-squares certificates.

THEOREM 1.3 (INFORMAL). *For every $\varepsilon > 0$, there is an $n^{O(1/\varepsilon)}$ time algorithm that takes input a bipartite graph $H = (U, V, E)$ with $|U| = k$, $|V| = n$ and each bipartite edge included in E with probability $p = 1/2$, and with probability at least 0.99 over the draw of H outputs an $n^{O(1/\varepsilon)}$ -time verifiable certificate that H contains no ℓ by k biclique for $\ell \leq n^\varepsilon$ whenever $k \geq \tilde{O}(\sqrt{n})$.*

Further, 1) the certificate can be expressed as an $O(1/\varepsilon)$ degree sum-of-squares refutation of the biclique axioms and 2) there does not exist a degree 2 certificate (equivalently, based on the “basic SDP”) to certify a bound of $\ell = o(n/k)$.

The results of prior works can be obtained by a simple spectral certificate captured by the basic SDP relaxation for upper bounding such bicliques (see detailed discussion in Section 2.1.1 of the techniques). In contrast, in this work, we depart from the spectral certificates and rely on a certain simple *geometric* certificate based on upper bounds on the size of sets of *pairwise negatively correlated vectors*. If $\varepsilon = O(\log \log n / \log n)$ is chosen such that $\ell = O(\log n)$ and $k = O(\sqrt{n \log n})$, the biclique refutation above translates into an algorithm for semi-random planted clique that works for $k \geq O(\sqrt{n \log n})$ in time $n^{O(\log n / \log \log n)}$, matching Steinhardt’s information-theoretic lower bound [52] and the threshold for the best known efficient algorithms for planted clique up to a $\sqrt{\log n}$ factor.

Hardness of refuting bicliques. We provide some evidence that improving on our biclique certification algorithms likely requires new techniques by proving a lower bound in the low-degree polynomial model (see [42] for a great exposition) is a restricted model for statistical *distinguishing*

problems. More precisely, the model considers problems where we are given a single sample (an instance of an algorithmic problem, a graph in our case) with the promise that it is an independent sample from one of two possible distributions: D_{null} — a distribution that does not admit solutions, usually a natural random model, and D_{planted} — a closely related distribution that does admit solutions. Informally speaking, the low-degree model restricts distinguishers to thresholds of low-degree polynomials of the input. While low-degree polynomials might appear restricted, they capture several algorithms including power iteration, approximate message passing, and local algorithms on graphs (cf. [15, 24]). Moreover, it turns out that they are enough to capture the best known spectral algorithms for several canonical problems such as planted clique, community detection, and sparse/tensor principal component analysis [8, 14, 28, 30].

This model arose naturally from work on constructing sum-of-squares lower bounds for the planted clique problem [8]. It was formalized in [28] and conjectured to imply sum-of-squares lower bounds for certain average-case refutation problems. Subsequently, starting with [30] (see also [27]), researchers have used the low-degree polynomial method as a technique to demarcate average-case algorithmic thresholds [24, 28, 51, 53].

In our case, D_{null} will be the $B(k, n, p)$ model: bipartite graphs with left vertex set of size k , right vertex set of size n , and each bipartite edge present in the graph with probability p . Notice that if we had an algorithm that certifies the absence of ℓ by $k - \ell$ bicliques in such a graph then we can distinguish between D_{null} and any D_{planted} supported on bipartite graphs that admit ℓ by $k - \ell$ bicliques. Thus the distinguishing problem is formally easier than the task of certification (also known as refutation). Despite the restrictedness of the low-degree model, we observe that for average-case planted clique with $k \gg \sqrt{n}$, constant degree polynomials suffice to distinguish between $D_{\text{null}} = G(n, 1/2)$ and $D_{\text{planted}} = G(n, 1/2) + k\text{-clique}$.

THEOREM 1.4 (LOW-DEGREE POLYNOMIAL HEURISTIC FOR BICLIQUE CERTIFICATION PROBLEM). *Fix $\varepsilon > 0$ small enough and n large enough. Let $D_{\text{null}} = B(k, n, 1/2)$ be the distribution on (k, n) -bipartite graphs where every edge is included independently with probability $1/2$. For $k = n^{1/2+\varepsilon}$, there is a distribution D_{planted} on (k, n) -bipartite graphs containing an ℓ by $k - \ell$ bipartite clique for $\ell = n^{0.1}$ such that the norm of the degree- $\Omega(1/\varepsilon)$ truncated likelihood ratio between D_{planted} and D_{null} is $1 + o(1)$.*

Informally speaking, the above theorem asserts that, for $k = n^{1/2+\varepsilon}$, statistical tests based on computing thresholds of $\Omega(1/\varepsilon)$ -degree polynomials fail to distinguish between D_{null} that does not admit $O(\log n)$ by k bicliques and D_{planted} that contains an $n^{0.1}$ by k biclique. It turns out that the most natural planted model (plant a random ℓ by $k - \ell$ clique and sample the rest of the graph independently) can be distinguished from D_{null} using just degree 1 polynomials and thus does not suffice to prove the above theorem. Instead, we use an edge-adjusted model where the probability of sampling edges outside the biclique is reduced in order to make the degree distribution of left vertices match that of $B(k, n, p)$.

For $p = 1/2$, the above theorem shows that we need polynomials of degree $O(1/\varepsilon)$ in order to distinguish between D_{null} and bipartite graphs with $n^{0.1}$ by $n^{1/2+\varepsilon}$ bicliques. Given the contrast to the

planted clique problem where the corresponding distinguishing problem can be solved by constant degree polynomials, we obtain some (weak) evidence that beating the guarantees of our current certificates may require new techniques for $p = 1/2$. For general p , a similar lower bound suggests that the degree of the polynomial required to distinguish between D_{null} and D_{planted} is larger than any function (independent of n) of $1/\varepsilon$, or else that k needs to scale with $1/(1 - p)$ instead of the information-theoretical optimal scaling of $1/\sqrt{1 - p}$ — note that this discrepancy is $\text{poly}(n)$ when $1 - p = 1/\text{poly}(n)$.

2 TECHNIQUES

In this section, we provide a high-level overview of our algorithm for the semi-random planted clique problem. For simplicity of exposition, we will focus on the important case of $p = 1/2$.

Given a graph G generated according to $\text{FK}(n, k, 1/2)$, our goal is to construct a small list of candidate k -cliques in G such that the true planted clique S^* is contained in the list (we will call such lists *correct*). Our construction will also ensure that a constant fraction of the vertices in S^* do not appear in any other clique in the list. As a result, we can also uniquely recover S^* with high probability when given, in addition, a uniformly random vertex in S^* .

Our algorithm and its analysis rely on the *proofs-to-algorithms* method (see [11, 23] for more on the usage of this method).

Inefficient algorithm. Let's first find an algorithm, even if inefficient, to generate a $\text{poly}(n)$ size correct list, i.e., one that contains S^* . Notice that simply outputting all k -cliques in G can lead to an exponentially large (i.e., $\sim n^k$) size list since we have no control over the subgraph induced on $[n] \setminus S^*$ (e.g., consider a clique on $[n] \setminus S^*$). Instead, we will enumerate all k -cliques in G that satisfy an additional property such that 1) the property is satisfied by the planted k -clique on S^* with high probability, and 2) every graph G has at most $(1 + o(1))n/k$ k -cliques satisfying the property. This property is quite natural and asks for the bipartite graph with the k -clique on the left and the rest of the vertices on the right to not contain a large *unbalanced* biclique with many vertices on the left side. Recall that an ℓ by r biclique in a bipartite graph H is a set of vertices that consists of ℓ left vertices and r right vertices such that H contains all possible bipartite edges between the two sides.

Definition 2.1 (Good k -cliques). Let G be a graph on n vertices. We say that a k -clique S in G is ℓ -good if every biclique (L, R) in the bipartite graph with left vertex set S , right vertex set $[n] \setminus S$, and edge set $\text{cut}_G(S)$ satisfies $|L| \leq \ell$ whenever $|R| \geq 1$ and $|L| + |R| = k$.

The planted k -clique on S^* is $O(\log n)$ -good with high probability over the draw of $\text{cut}(S^*)$.

PROPOSITION 2.2 (BIPARTITE CLIQUE NUMBER OF $\text{cut}(S^*)$). *Let $k, n \in \mathbb{N}$ and $G \sim \text{FK}(n, k, 1/2)$. Then, for large enough n and a constant $c > 0$, with probability at least 0.99 over the draw of edges in $\text{cut}(S^*)$, for any $L \subseteq S^*$, $R \subseteq [n] \setminus S^*$ such that (L, R) is a biclique in $\text{cut}(S^*)$ satisfying $|R| \geq 1$ and $|L| + |R| = k$, we have $|L| \leq c \log_2 n$.*

PROOF. The proof is a simple application of the first moment method. Note that it is enough to argue the proposition in the absence of the monotone adversary as deleting any subset of edges in $\text{cut}(S^*)$ maintains the goodness of S^* .

The probability that $\text{cut}(S^*)$ contains all the edges between $L \subseteq S^*$ and $R \subseteq [n] \setminus S^*$ is at most $2^{-(k-|L|)|L|}$. Thus, the expected number of bicliques (L, R) such that $|L| \geq c \log_2 n$ is at most $\sum_{\ell+r=k, \ell \geq c \log_2 n} \binom{n-k}{k-\ell} \binom{k}{\ell} 2^{-(k-\ell)\ell} \rightarrow 0$ as $n \rightarrow \infty$ if c is a large enough constant. The proposition then follows by an application of Markov's inequality. $\square$

A simple greedy argument upper bounds the number of ℓ -good k -cliques if $k \geq O(\sqrt{n \ell \log n})$.

PROPOSITION 2.3 (NUMBER OF GOOD k -CLIQUEs). *Let G be a graph on n vertices. Then, for any ℓ , if $k > 2\sqrt{n \ell / \delta}$ for some $\delta < 1$, then the number of ℓ -good k -cliques in G is at most $(1 + \delta)n/k$.*

PROOF. Suppose not and take any $m = (1 + \delta)n/k$ such good k -cliques. Observe that any pair of ℓ -good k -cliques S, S' can only intersect in at most ℓ vertices, as otherwise $\text{cut}(S)$ would contain a biclique with more than ℓ left vertices. Thus, the m good k -cliques must cover at least $mk - m^2 \ell = n + \delta n - (4n^2/k^2)\ell$ vertices, a number that exceeds the total number of vertices n if $k > 2\sqrt{n \ell / \delta}$. $\square$

Propositions 2.2 and 2.3 immediately yield an $n^{O(k)}$ time algorithm to generate a correct list of k -cliques of size $(1 + \delta)(n/k)$. In fact, this algorithm can be made to run in time $n^{O(\log n)}$ by enumerating all $c \log_2 n$ size cliques Q in G and adding a k -clique to the list if the common neighborhood of Q is of size $\geq k - |Q|$ and forms a clique with Q .

2.1 Efficient Algorithms and Biclique Certificates

In the inefficient algorithm above a key idea is the claim that $\text{cut}(S^*)$ does not have an ℓ by $k - \ell$ bipartite clique for $\ell > O(\log n)$. Note that $\text{cut}(S^*)$ is an unbalanced (left side is much smaller than the right) k by $n - k \approx n$ bipartite graph and we proved that it does not have an unbalanced ($\gg O(\log n)$ vertices from the left) biclique in it.

Key to our efficient algorithm for semi-random planted clique is an efficiently computable *certificate* of non-existence of unbalanced bicliques in H as above (i.e., a *refutation*).

Let $B(n_1, n_2, p)$ denote the distribution on bipartite graphs with n_1 left and n_2 right vertices and every bipartite edge included with probability p independently. Let us phrase the version relevant to us formally before continuing:

Definition 2.4 (Refuting unbalanced bicliques). An algorithm that takes as input a bipartite graph $H = (U, V, E)$ with $|U| = k$, $|V| = n - k$ refutes ℓ by $k - \ell$ bicliques in random k by $n - k$ bipartite graphs if it has the following two properties:

- (1) **Correctness:** If the algorithm outputs s , then there is no s by $k - s$ biclique in H .
- (2) **Utility:** If $H \sim B(k, n - k, 1/2)$, then the algorithm outputs $s \leq \ell$ with probability at least 0.99 over the draw of H .

Remark 2.5 (From certificates to algorithms: a heuristic). In Section 2.3, we overview the translation of a (constant degree sum-of-squares) certificate that the left side of any size- k biclique in $H \sim B(k, n - k, 1/2)$ has at most ℓ vertices into an algorithm for the semi-random planted k -clique problem that succeeds whenever

$k \geq O(\sqrt{n \ell})$. This matches the simple bound in Proposition 2.3 for the “brute-force” algorithm above. We postpone the discussion of sum-of-squares proofs for now while noting that all certificates discussed in this section are in fact constant degree sum-of-squares certificates.

Observe that our simple analysis of the inefficient algorithm gives an $n^{O(\log n)}$ algorithm that refutes the existence of ℓ by $k - \ell$ bicliques in $B(k, n - k, 1/2)$ with probability at least 0.99 for $\ell = O(\log n)$. Our goal is to find a polynomial time algorithm that succeeds for ℓ as close to $O(\log n)$ as possible.

The biclique refutation problem appears to be an interesting analog of refuting cliques in random (non-bipartite) graphs $G \sim G(n, 1/2)$ (that underlies algorithms for the fully-random planted clique problem) or bicliques in $B(n, n, 1/2)$ (i.e., the balanced bipartite graph). It can be thought of as *certifying* the correctness of the candidates in the list that is purportedly a solution to the semi-random planted clique problem. Finding solutions together with a certificate of correctness is an important goal by itself. For example, this is a key advantage (in addition to tolerating a monotone adversary) of the method of Feige and Krauthgamer [19] over the spectral algorithm [1] for the planted clique problem.

2.1.1 Basic Spectral Certificate. Let us start by recalling the basic spectral certificate that underlies the algorithms for the average-case planted clique problem. This certificate implicitly underlies the algorithms of [13, 45]. Our framework translates it into an algorithm for semi-random planted clique whenever $k \gg O(n^{2/3})$.

PROPOSITION 2.6 (BASIC SPECTRAL CERTIFICATE FOR CLIQUE NUMBER). *In any graph G , the clique number $\omega(G) \leq 1 + \|A\|_2$ where A is the $\{\pm 1\}$ adjacency matrix of G .*

PROOF. If x is a $\{0, 1\}$ -indicator of a k -clique in G , then, note that $k(k - 1) = x^T A x \leq \|x\|_2^2 \|A\|_2 = k \|A\|_2$. Thus, $k \leq 1 + \|A\|_2$ for any graph G . $\square$

Thus, simply outputting the (polynomial time computable) largest singular value of A gives a certificate of an upper bound on $\omega(G)$. Further, if $G \sim G(n, 1/2)$, then standard spectral norm bounds on random symmetric $\{\pm 1\}$ matrices imply that the algorithm outputs with high probability a bound of $O(\sqrt{n})$.

Let's now see an analog of this method for bicliques.

PROPOSITION 2.7 (BASIC SPECTRAL CERTIFICATE FOR BICLIQUES). *Let H be the $\{\pm 1\}$ adjacency matrix of a k by $n - k$ bipartite graph H . For any k -clique in H , the number of left vertices ℓ satisfies $\ell(k - \ell) \leq \|H\|_2^2$.*

PROOF. Let x, y be the $\{0, 1\}$ indicators of the left and right sides of a biclique in H . Then, $\|x\|_2^2 \|y\|_2^2 = x^T H y \leq \|x\|_2 \|y\|_2 \|H\|_2$. Or, $(\sum_i x_i)(\sum_i y_i) = \|x\|_2 \|y\|_2 \leq \|H\|_2^2$. $\square$

For a random bipartite graph from $B(k, n - k, 1/2)$, the H is a k by $n - k$ matrix with independent random $\{\pm 1\}$ entries. For such matrices, standard results show that $\|H\|_2 \leq O(\sqrt{k} + \sqrt{n}) = O(\sqrt{n})$. Further, by a union bound, the degrees of all right vertices are at most $k/2 + O(\sqrt{k \log n})$ with high probability, so $k - \ell \geq k/4$ if $k \gg \log n$. In that case, the above proposition shows that the

spectral certificate refutes the existence of an ℓ by $k - \ell$ clique for $\ell \leq O(n/k)$.

By applying the heuristic from Remark 2.5, we obtain an algorithm for semi-random planted clique if $k \geq O(\sqrt{n\ell})$ with $\ell = O(n/k)$, that is, if $k \geq O(n^{2/3})$, matching the guarantees of [45].

It turns out that the bound of $\ell = O(n/k)$ based on the basic SDP/spectral relaxations is essentially tight. We show that the basic SDP provably fails to certify that $\ell = o(n/k)$. This shows an inherent limitation of certificates based on the basic SDP/spectral relaxations.

The Charikar-Steinhardt-Valiant approach. In their work on algorithms for list-decodable mean estimation [13], the authors devised a method for the analog of the semi-random planted clique problem without the monotone adversary step. When viewed from our vantage point of biclique refutation, their idea can be thought of as taking the ± 1 -neighborhood indicators of the *right* hand side of the graph and treating them as $n - k$ samples of a k -dimensional distribution. An ℓ by $k - \ell$ biclique translates¹ into the distribution having a non-zero mean. Thus, one can apply (analog of) list-decodable mean estimation algorithms [13, 39] to refute the existence of bicliques. The guarantees of the algorithm depend on higher directional moments of the input distribution. The “base case” corresponds to using just the second moments of the distribution – and this roughly relates to the use of the basic spectral certificate above. The higher moment variants can indeed yield improvements but this does not apply to our setting, because when seen from the vantage point of list-decodable mean estimation we have $n \ll k^2$ samples of a k -dimensional distribution – a bound not sufficient for the 4th moments to converge! Indeed, this is the key bottleneck that leads to a barrier at $k = \tilde{O}(n^{2/3})$ for the CSV approach (and led to Steinhardt’s open question for semi-random planted clique [52]).

2.1.2 Improved Spectral Certificates. Can we improve on the basic spectral certificate? We note that for related problems (e.g., densest k -subgraph, random constraint satisfaction, coloring random graphs) we usually get no asymptotic improvement by considering spectral certificates with larger (but polynomial size) matrices built from the instance. Indeed, one can prove strong lower bounds [33, 38] that rule out such larger polynomial size certificates captured by constant degree sum-of-squares proofs.

Neighborhood reduction. A natural way to improve the spectral certificate for the clique number of $G \sim G(n, 1/2)$ from Proposition 2.6 is to cycle through all possible subsets of t vertices, move to the common neighborhood of the t vertices and then apply Proposition 2.6 to the induced graph on this common neighborhood. This strategy yields an upper bound of $\omega(G) \leq t + 1 + \max_{S \subseteq [n], |S|=t} \|A_S\|_2$ where A_S is the adjacency matrix of the induced subgraph on the common neighborhood of S . One can prove that $\|A_S\|_2 \leq O(\sqrt{n/2^t})$ with high probability simultaneously for all S of size t , certifying an upper bound of $O(\sqrt{n/2^t})$ on the clique number $\omega(G)$. Since the resulting certificate has polynomial size

¹We note that the CSV approach directly applies to the semi-random planted clique model and does not actually yield a biclique certificate. The reason is that an ℓ by $k - \ell$ biclique does not translate into non-zero mean for arbitrary bipartite graphs. We ignore this distinction in order to allow an intuitive comparison of their technique in the context of our work.

only when $t = O(1)$, the improvement makes no asymptotic difference in the threshold k at which polynomial time algorithms work. As an aside, this simple certificate happens to be optimal for the degree t Lovász-Schrijver SDP hierarchy [20] applied to $G \sim G(n, 1/2)$. Repeating an analogous argument in our case also yields no asymptotic improvement unless $t = \omega(1)$ (though it does allow us to get arbitrary constant factor improvements).

Tensoring. We consider next a class of “tensoring” schemes for producing improved spectral certificates. Consider a bipartite graph with $\{\pm 1\}$ adjacency matrix H' with the same right side but the left side containing all pairs of left vertices from H . The $((i, j), k)$ -th entry of H' equals $H(i, k)H(j, k)$ – the “parity” or product of the $\{\pm 1\}$ indicators of edges (i, k) and (j, k) in H . H' is a k^2 by n matrix, and further, an ℓ by $k - \ell$ biclique in H translates into an ℓ^2 by $k - \ell$ biclique in H' . The basic spectral certificate applied to H' yields that $\ell^2 \leq O(\|H'\|_2^2/k)$.

If H' were a matrix of independent random $\{\pm 1\}$ entries, $\|H'\|_2 = O(\sqrt{k^2}) = O(k)$ yielding $\ell \leq O(\sqrt{k})$. Despite H' having correlations in its entries, this optimistic² bound is essentially correct (we will omit the proof here). Plugging this back into our heuristic, we get an algorithm for semi-random planted k -clique if $k \geq O(\sqrt{n\sqrt{k}})$ or $k \gg n^{2/3}$, the same as before! That is, even though the tensoring trick gives a different asymptotic estimate, it does not lead to any improvement in the threshold for k in our semi-random planted clique application.

What happens if we “tensor the left side” t times for $t > 2$? An optimistic estimate such as the above yields a bound of $\ell^t \leq O(k^{t-1})$ or $\ell \leq k^{1-1/t}$ – a bound that appears to *degrade* as we increase t ! We will omit the details here but a similarly worse bound results if we tensor the right side of H instead.

Two-sided tensoring beats the $n^{2/3}$ barrier but fails a long way off $\sqrt{n}$. It turns out simultaneously tensoring both sides unequally helps beat the $\ell \leq \max\{\sqrt{k}, n/k\}$ bound obtained via one-sided tensoring above. Intuitively speaking, the “optimal” two-sided tensoring attempts to make the resulting adjacency matrix as “square” in dimensions as possible. Formal proofs require analyzing matrices of correlated random entries using the graph matrix method devised in the context of proving sum-of-squares lower bounds in [9] and follow-ups. We note without further details that two-sided tensoring appears to break down at $k \sim n^{0.61}$.

2.2 Our Certificate: Bicliques Imply Sets of Negatively Correlated Vectors

Our key idea to circumvent the bottlenecks in the natural spectral certificates is to abandon the idea of spectral certificates altogether. Instead, we will show that a simple family of “geometric” certificates for biclique numbers allows us to show $\ell \leq n^\epsilon$ for any fixed $\epsilon > 0$. Specifically, we will show that if there is an ℓ by $k - \ell$ biclique in H , then one can extract $2^\ell - 1$ *pairwise negatively correlated vectors* in n dimensions.

In order to explain this connection, let us note a property of a random bipartite graph $H = (U, V, E) \sim B(k, n - k, 1/2)$. For any

²Every rectangular matrix of larger dimension k^2 and Frobenius norm $k\sqrt{n}$ has a spectral norm $\geq k$.

subset $S \subseteq U$ of $|S| \leq t$ vertices from the left vertex set of H , let $N_S(j) = \prod_{i \in S} H(i, j)$ where H is the $\{\pm 1\}$ -adjacency matrix of H . Then N_S is an n dimensional vector of “parities” of $\{\pm 1\}$ indicators of all edges from S to $\{k\}$. Further, in a random H , every N_S is nearly *balanced*. That is, by a simple Chernoff and union bound argument, $|\sum_{i \leq n-k} N_S(i)| \leq O(\sqrt{nt \log n})$ for every S of size t .

Let’s call a k by $n-k$ bipartite graph t -fold balanced if the above property holds: that is, every N_S is approximately balanced for $|S| \leq t$. We will now show that given an ℓ by $k-\ell$ biclique in a t -fold balanced graph, we can produce a set of $\binom{\ell}{t/2}$ pairwise negatively correlated vectors in n dimensions.

PROPOSITION 2.8 (BICLIQUES AND NEGATIVELY CORRELATED VECTORS). *Suppose H is a k by $n-k$ bipartite graph that is t -fold balanced for some $t \in \mathbb{N}$. Suppose that H contains an ℓ by $k-\ell$ biclique (L, R) for $k-\ell \geq k/4$. Then, if $k \geq O(\sqrt{nt \log n})$, there exist $\binom{\ell}{t/2}$ different $(n-2k+\ell)$ -dimensional vectors N_S^- (one for each $S \subseteq L$ of size $t/2$) such that $\langle N_S^-, N_T^- \rangle < 0$ whenever $S \neq T$.*

PROOF. First observe that for any $S, T \subseteq L$ of size $t/2$, $\langle N_S, N_T \rangle = \sum_{j \leq n-k} N_{S \Delta T}(j) = O(\sqrt{nt \log n})$ where we invoked the t -fold balancedness of H . Now, without loss of generality, assume that R is the set of the first $k-\ell$ vertices on the right. Consider the vectors N_S^- in $n-2k+\ell$ dimensions obtained by stripping the first $k-\ell$ coordinates off of N_S for every $S \subseteq L$ of size $t/2$. Since $S, T \subseteq L$, the first $k-\ell$ coordinates contribute $+(k-\ell)$ to $\langle N_S, N_T \rangle$. Thus, $\langle N_S^-, N_T^- \rangle \leq O(\sqrt{nt \log n}) - k/4 < 0$ if $k-\ell \geq k/4$ and $k \geq O(\sqrt{nt \log n})$. $\square$

It is a standard fact that there can only be $d+1$ pairwise negatively correlated vectors in d dimensions. A weaker version can be proved via a simple argument involving quadratic polynomials over the vectors:

PROPOSITION 2.9 (BOUND ON NEGATIVELY CORRELATED VECTORS). *Let $v_1, v_2, \dots, v_N$ be n -dimensional vectors of length $\sqrt{n}$ each satisfying $\langle v_i, v_j \rangle \leq -r$. Then $N \leq 1 + n/r$.*

PROOF. We know that $\|\sum_{i \leq N} v_i\|_2^2 \geq 0$. On the other hand, $\|\sum_{i=1}^N v_i\|_2^2 = \sum_{i=1}^N \|v_i\|_2^2 + \sum_{i \neq j} \langle v_i, v_j \rangle \leq Nn - N(N-1)r$. Putting the lower and upper bound together yields that $N-1 \leq n/r$ or $N \leq 1 + n/r$. $\square$

Now, Proposition 2.8 yields $\binom{\ell}{t/2}$ vectors with pairwise correlations at most $-ck$ for some constant $c > 0$ if $k \gg \sqrt{nt \log n}$. On the other hand, Proposition 2.9 yields that the number of such vectors can only be $1 + O(n/k)$. Putting these two bounds together yields that $\ell \lesssim (n/k)^{2/t}$. Choosing $t = 1/\epsilon$ gives us an $n^{O(1/\epsilon)}$ size certificate that ℓ is at most n^ϵ .

The above argument can be converted into a sum-of-squares refutation of bicliques in H . The main observation is that the step where we strip the first $k-\ell$ coordinates off of N_S can be done “within sum-of-squares” while the remaining argument is a sum-of-squares proof by virtue of the above simple proposition. It turns out that we need some additional careful arguments to place the certificate in a usable form, which we will omit for the purpose of this overview.

2.3 From Biclique Certificates to Algorithms for Semi-random Planted Clique

Our algorithms use the biclique certificates discussed previously to analyze a rounding algorithm for SDP relaxations of the standard k -clique axioms. Specifically, consider the standard integer programming formulation of the k -clique problem written as the quadratic polynomial system $\mathcal{A} = \mathcal{A}(G)$ below. Note that the solutions to $\mathcal{A}(G)$ are k -cliques in the graph G on vertex set $[n]$.

$$\mathcal{A}(G): \left\{ \begin{array}{ll} \forall i \in [n] & w_i^2 = w_i \\ \forall i \in [n] & \sum_{j=1}^n w_i w_j = k \\ \forall i, j \text{ s.t. } \{i, j\} \notin G & w_i w_j = 0 \end{array} \right\} \quad (1)$$

Finding a solution to this quadratic program is clearly NP-hard. So we will instead work with “sum-of-squares” SDP relaxations of the quadratic program, whose solutions can be interpreted as a generalization of probability distributions over solutions to the quadratic program. Specifically, a degree d pseudo-distribution D is a relaxation of a probability distribution on $\{0, 1\}^n$ in that the associated “mass” function can take negative values while still inheriting a non-trivial subset of the properties of probability distributions. For now note the following relevant bits: 1) Unlike an actual probability distribution, we only get access to low-degree moments (i.e., expectations of monomials) of D and thus can only compute expectations of degree $\leq d$ polynomials, 2) pseudo-distributions can assign “negative probabilities” and thus may not assign non-negative expectations to pointwise non-negative degree d polynomials f , but 3) degree d pseudo-distributions do assign non-negative expectations to any f that is a sum of squares of degree $\leq d/2$ polynomials, and 4) a pseudo-distribution of degree d satisfying $\mathcal{A}$ satisfies all “low-degree inferable” properties of k -cliques but need not be supported on w that indicate k -cliques at all. Here, low-degree inferable property means that for any degree $\leq d-2$ polynomial f and any $\{i, j\} \notin G$, $\mathbb{E}_D[f w_i w_j] = 0$.

A degree d pseudo-distribution minimizing any convex objective in the pseudomoments $\mathbb{E}[\prod_{i \in S} w_i]$ for $|S| \leq d$ and approximately satisfying $\mathcal{A}$ at degree d can be computed in time $n^{O(d)}$.

Though a pseudo-distribution is not a probability distribution over solutions to $\mathcal{A}$, it is still helpful for the reader to imagine it to be as such.

How do our biclique certificates help us? It turns out that while degree d pseudo-distributions are far from actual probability distributions for $d \ll n$, they behave so for the purpose of polynomial inequalities that can be derived from $\mathcal{A}$ using degree d sum-of-squares proofs. The conclusion of our biclique certificate from Proposition 2.2 can be written as a degree $O(t)$ consequence of the quadratic system $\mathcal{B}$ that identifies bicliques in bipartite graphs of total size k :

$$\mathcal{B}(H): \left\{ \begin{array}{ll} \forall u \in U & x_u^2 = x_u \\ \forall v \in V & y_v^2 = y_v \\ & |x| + |y| = k \\ \forall u \in U, v \in V \text{ s.t. } \{u, v\} \notin E & x_u y_v = 0 \end{array} \right\}. \quad (2)$$

Consider the bipartite graph cut (S^*) . Let w_L be the restriction of w to coordinates in S^* and w_R be the restriction of w to coordinates outside of S^* . Then, $\mathcal{A}$ implies that (w_L, w_R) satisfy $\mathcal{B}$ for the

bipartite graph cut(S^*). Since the pseudo-distribution D satisfies $\mathcal{A}$, we can conclude that

$$\tilde{\mathbb{E}}_D \left[\left(\sum_{i \in S^*} w_i \right)^t \left(\sum_{i \notin S^*} w_i \right) \right] \leq O(n^5/k^4) \quad (3)$$

whenever the pseudo-distribution D has degree at least $O(t)$. Note that S^* is not known to us but the above inequality forces the pseudo-distribution computed by the SDP to capture some non-trivial information about it.

The need for coverage constraints. Roughly speaking, (3) can be interpreted as saying that the pseudo-distribution is “supported” only on those w that cannot simultaneously appreciably intersect S^* and $[n] \setminus S^*$. Such a fact by itself seems unhelpful. After all, the pseudo-distribution could completely ignore S^* and focus on the “worst-case” graph on $[n] \setminus S^*$. Given the worst-case hardness of clique, the pseudo-distribution may not have any information about k -cliques in $[n] \setminus S^*$ and consequently the input graph.

In order to make (3) useful, we must somehow “force” the pseudo-distribution to have a non-trivial mass on vertices in S^* . Of course, we do not know S^* , so how can we do it? It turns out that this can be accomplished by certain “max coverage” constraints. Specifically, instead of finding any pseudo-distribution consistent with $\mathcal{A}$, we find one that minimizes $\|\tilde{\mathbb{E}}_D[w]\|_2^2$. This is a convex function of the pseudo-distribution and thus can be minimized efficiently using the ellipsoid method. This objective forces the pseudo-distribution to be “spread-out”. Indeed, in a different language, such an objective is used also in [45], though arguably our treatment of such an objective as a max coverage constraint on sum-of-squares relaxations of $\mathcal{A}$ appears to demystify the use of crude-SDP in [45]. We note that such a max coverage constraint is at the heart of the rounding algorithms for several problems in list-decodable learning starting with [34].

A key consequence of the max coverage constraint is that, by an elementary convexity argument, it implies the following proposition:

PROPOSITION 2.10 (MAX COVERAGE PSEUDO-DISTRIBUTIONS). *For any pseudo-distribution D on w satisfying $\mathcal{A}$ of degree at least 2 and minimizing $\|\tilde{\mathbb{E}}_D[w]\|_2^2$, we have $\sum_{i \in S^*} \tilde{\mathbb{E}}_D[w_i] \geq \frac{k^2}{n}$.*

A rounding algorithm now falls naturally out of the above two discussions. We look at an n^t by n matrix indexed by subsets of size $t = O(1/\epsilon)$ on the rows and singleton vertices on the columns, whose value at index (S, i) is $\frac{\tilde{\mathbb{E}}_D[w_S w_i]}{\tilde{\mathbb{E}}_D[w_S]}$. Proposition 2.10 implies that the rows of this (huge) matrix corresponding to the unknown planted clique must have a large total sum. On the other hand, as a consequence of the biclique certificate, we learn that for such rows the columns corresponding to $[n] \setminus S^*$ must have a low total contribution. Together these two statements allow us to use a simple greedy algorithm that selects a uniformly random row of the above matrix and takes the largest $\sim k$ entries to recover a list containing a set of $\sim k$ vertices that has a large constant fraction intersection with S^* . Such a set can then be refined using a simple combinatorial “cleanup” step.

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REFERENCES

- [1] Noga Alon, Michael Krivelevich, and Benny Sudakov. 1998. Finding a large hidden clique in a random graph. In *Proceedings of the Eighth International Conference “Random Structures and Algorithms”* (Poznan, 1997). *Random Structures Algorithms* 13, 3-4, 457–466. [https://doi.org/10.1002/\(SICI\)1098-2418\(199810/12\)13:3/4<457::AID-RSA14>3.3.CO;2-K](https://doi.org/10.1002/(SICI)1098-2418(199810/12)13:3/4<457::AID-RSA14>3.3.CO;2-K)
- [2] Sanjeev Arora, Boaz Barak, and David Steurer. 2015. Subexponential algorithms for unique games and related problems. *J. ACM* 62, 5 (2015), Art. 42, 25. <https://doi.org/10.1145/2775105>
- [3] Ainesh Bakshi, Ilias Diakonikolas, Samuel B. Hopkins, Daniel Kane, Sushrut Karmalkar, and Pravesh K. Kothari. 2020. Outlier-Robust Clustering of Gaussians and Other Non-Spherical Mixtures. In *61st IEEE Annual Symposium on Foundations of Computer Science, FOCS 2020, Durham, NC, USA, November 16–19, 2020*, Sandy Irani (Ed.). IEEE, 149–159. <https://doi.org/10.1109/FOCS46700.2020.00023>
- [4] Ainesh Bakshi, Ilias Diakonikolas, He Jia, Daniel M. Kane, Pravesh K. Kothari, and Santosh S. Vempala. 2020. Robustly Learning Mixtures of k Arbitrary Gaussians. *CoRR* abs/2012.02119 (2020). [arXiv:2012.02119](https://arxiv.org/abs/2012.02119) <https://arxiv.org/abs/2012.02119>
- [5] Ainesh Bakshi and Pravesh Kothari. 2020. Outlier-Robust Clustering of Non-Spherical Mixtures. *CoRR* abs/2005.02970 (2020). [arXiv:2005.02970](https://arxiv.org/abs/2005.02970) <https://arxiv.org/abs/2005.02970>
- [6] Ainesh Bakshi and Pravesh K. Kothari. 2021. List-Decodable Subspace Recovery: Dimension Independent Error in Polynomial Time. In *Proceedings of the 2021 ACM-SIAM Symposium on Discrete Algorithms, SODA 2021, Virtual Conference, January 10 – 13, 2021*, Daniel Marx (Ed.). SIAM, 1279–1297. <https://doi.org/10.1137/1.9781611976465.78>
- [7] Ainesh Bakshi and Adarsh Prasad. 2021. Robust linear regression: optimal rates in polynomial time. In *STOC ’21: 53rd Annual ACM SIGACT Symposium on Theory of Computing, Virtual Event, Italy, June 21–25, 2021*, Samir Khuller and Virginia Vassilevska Williams (Eds.). ACM, 102–115. <https://doi.org/10.1145/3406325.3451001>
- [8] Boaz Barak, Samuel Hopkins, Jonathan Kelner, Pravesh K Kothari, Ankur Moitra, and Aaron Potechin. 2019. A nearly tight sum-of-squares lower bound for the planted clique problem. *SIAM J. Comput.* 48, 2 (2019), 687–735.
- [9] Boaz Barak, Samuel B. Hopkins, Jonathan A. Kelner, Pravesh Kothari, Ankur Moitra, and Aaron Potechin. 2016. A Nearly Tight Sum-of-Squares Lower Bound for the Planted Clique Problem. In *FOCS. IEEE Computer Society*, 428–437.
- [10] Boaz Barak, Prasad Raghavendra, and David Steurer. 2011. Rounding semidefinite programming hierarchies via global correlation. In *2011 IEEE 52nd Annual Symposium on Foundations of Computer Science—FOCS 2011*. IEEE Computer Soc., Los Alamitos, CA, 472–481. <https://doi.org/10.1109/FOCS.2011.95>
- [11] Boaz Barak and David Steurer. 2016. Proofs, beliefs, and algorithms through the lens of sum-of-squares. Lecture notes in preparation, available on <http://sumofsquares.org..>
- [12] A. Blum and J. Spencer. 1995. Coloring Random and Semi-Random k -Colorable Graphs. *Journal of Algorithms* 19, 2 (1995), 204 – 234. <https://doi.org/10.1006/jagm.1995.1034>
- [13] Moses Charikar, Jacob Steinhardt, and Gregory Valiant. 2017. Learning from untrusted data. In *STOC. ACM*, 47–60.
- [14] Yunzi Ding, Dmitry Kunisky, Alexander S Wein, and Afonso S Bandeira. 2019. Subexponential-time algorithms for sparse PCA. *arXiv preprint arXiv:1907.11635* (2019).
- [15] David L Donoho, Arian Maleki, and Andrea Montanari. 2009. Message-passing algorithms for compressed sensing. *Proceedings of the National Academy of Sciences* 106, 45 (2009), 18914–18919.
- [16] Uriel Feige. 2019. Introduction to Semirandom Models. In *Beyond Worst-case Analysis of Algorithms*, Tim Roughgarden (Ed.). Oxford, Chapter 10, 266–290.
- [17] Uriel Feige and Joe Kilian. 1998. Heuristics for finding large independent sets, with applications to coloring semi-random graphs. In *Proceedings 39th Annual Symposium on Foundations of Computer Science (Cat. No. 98CB36280)*. IEEE, 674–683.
- [18] Uriel Feige and Joe Kilian. 2001. Heuristics for Semirandom Graph Problems. *J. Comput. System Sci.* 63, 4 (2001), 639 – 671. <https://doi.org/10.1006/jcss.2001.1773>
- [19] Uriel Feige and Robert Krauthgamer. 2000. Finding and certifying a large hidden clique in a semirandom graph. *Random Structures Algorithms* 16, 2 (2000), 195–208. [https://doi.org/10.1002/\(SICI\)1098-2418\(200003\)16:2<195::AID-RSA5>3.3.CO;2-1](https://doi.org/10.1002/(SICI)1098-2418(200003)16:2<195::AID-RSA5>3.3.CO;2-1)

- [20] Uriel Feige and Robert Krauthgamer. 2003. The probable value of the Lovász-Schrijver relaxations for maximum independent set. *SIAM J. Comput.* 32, 2 (2003), 345–370. <https://doi.org/10.1137/S009753970240118X>
- [21] Vitaly Feldman, Elena Grigorescu, Lev Reyzin, Santosh Vempala, and Ying Xiao. 2013. Statistical algorithms and a lower bound for detecting planted cliques. In *STOC*. ACM, 655–664.
- [22] Vitaly Feldman, Elena Grigorescu, Lev Reyzin, Santosh S. Vempala, and Ying Xiao. 2017. Statistical algorithms and a lower bound for detecting planted cliques. *J. ACM* 64, 2 (2017), Art. 8, 37. <https://doi.org/10.1145/3046674>
- [23] Noah Fleming, Pravesh Kothari, and Toniann Pitassi. 2019. Semialgebraic Proofs and Efficient Algorithm Design. *Foundations and Trends® in Theoretical Computer Science* 14, 1–2 (2019), 1–221. <https://doi.org/10.1561/04000000086>
- [24] David Gamarnik, Aukosh Jagannath, and Alexander S Wein. 2020. Low-degree hardness of random optimization problems. *arXiv preprint arXiv:2004.12063* (2020).
- [25] Michel X. Goemans and David P. Williamson. 1995. Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming. *J. Assoc. Comput. Mach.* 42, 6 (1995), 1115–1145. <https://doi.org/10.1145/227683.227684>
- [26] Johan Håstad. 1999. Clique is hard to approximate within $n^{1-\epsilon}$. *Acta Math.* 182, 1 (1999), 105–142. <https://doi.org/10.1007/BF02392825>
- [27] Samuel Hopkins. 2018. Statistical inference and the sum of squares method. *PhD thesis, Cornell University* (2018).
- [28] Samuel B Hopkins, Pravesh K Kothari, Aaron Potechin, Prasad Raghavendra, Tselil Schramm, and David Steurer. 2017. The power of sum-of-squares for detecting hidden structures. In *2017 IEEE 58th Annual Symposium on Foundations of Computer Science (FOCS)*. IEEE, 720–731.
- [29] Samuel B. Hopkins and Jerry Li. 2018. Mixture models, robustness, and sum of squares proofs. In *Proceedings of the 50th Annual ACM SIGACT Symposium on Theory of Computing, STOC 2018, Los Angeles, CA, USA, June 25–29, 2018*, Ilias Diakonikolas, David Kempe, and Monika Henzinger (Eds.). ACM, 1021–1034. <https://doi.org/10.1145/3188745.3188748>
- [30] Samuel B Hopkins and David Steurer. 2017. Efficient bayesian estimation from few samples: community detection and related problems. In *2017 IEEE 58th Annual Symposium on Foundations of Computer Science (FOCS)*. IEEE, 379–390.
- [31] Misha Ivkov and Pravesh K. Kothari. 2022. List-decodable covariance estimation. In *STOC '22: 54th Annual ACM SIGACT Symposium on Theory of Computing, Rome, Italy, June 20 - 24, 2022*, Stefano Leonardi and Anupam Gupta (Eds.). ACM, 1276–1283. <https://doi.org/10.1145/3519935.3520006>
- [32] Mark Jerrum. 1992. Large Cliques Elude the Metropolis Process. *Random Struct. Algorithms* 3, 4 (1992), 347–360.
- [33] Chris Jones, Aaron Potechin, Goutham Rajendran, Madhur Tulsiani, and Jeff Xu. [2022] ©2022. Sum-of-squares lower bounds for sparse independent set. In *2021 IEEE 62nd Annual Symposium on Foundations of Computer Science—FOCS 2021*. IEEE Computer Soc., Los Alamitos, CA, 406–416.
- [34] Sushrut Karmalkar, Adam R. Klivans, and Pravesh Kothari. 2019. List-decodable Linear Regression. In *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8–14, 2019, Vancouver, BC, Canada*, Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d'Alché-Buc, Emily B. Fox, and Roman Garnett (Eds.). 7423–7432. <https://proceedings.neurips.cc/paper/2019/hash/7f5fc754c7af0a6370c9bf91314e79f4-Abstract.html>
- [35] Richard M Karp. 1972. Reducibility among combinatorial problems. In *Complexity of computer computations*. Springer, 85–103.
- [36] Subhash Khot. 2014. Hardness of approximation. In *Proceedings of the International Congress of Mathematicians—Seoul 2014. Vol. 1*. Kyung Moon Sa, Seoul, 711–728.
- [37] Adam R. Klivans, Pravesh K. Kothari, and Raghu Meka. 2018. Efficient Algorithms for Outlier-Robust Regression. In *Conference On Learning Theory, COLT 2018, Stockholm, Sweden, 6–9 July 2018*. 1420–1430. <http://proceedings.mlr.press/v75/klivans18a.html>
- [38] Pravesh K. Kothari, Ryuhei Mori, Ryan O'Donnell, and David Witmer. 2017. Sum of squares lower bounds for refuting any CSP. In *STOC*. ACM, 132–145.
- [39] Pravesh K. Kothari and Jacob Steinhardt. 2017. Better Agnostic Clustering via Relaxed Tensor Norms. (2017).
- [40] Pravesh K. Kothari, Jacob Steinhardt, and David Steurer. 2018. Robust moment estimation and improved clustering via sum of squares. In *Proceedings of the 50th Annual ACM SIGACT Symposium on Theory of Computing, STOC 2018, Los Angeles, CA, USA, June 25–29, 2018*, Ilias Diakonikolas, David Kempe, and Monika Henzinger (Eds.). ACM, 1035–1046. <https://doi.org/10.1145/3188745.3188970>
- [41] Ludek Kucera. 1995. Expected Complexity of Graph Partitioning Problems. *Discrete Applied Mathematics* 57, 2–3 (1995), 193–212.
- [42] Dmitriy Kunisky, Alexander S Wein, and Afonso S Bandeira. 2019. Notes on computational hardness of hypothesis testing: Predictions using the low-degree likelihood ratio. *arXiv preprint arXiv:1907.11636* (2019).
- [43] Allen Liu and Ankur Moitra. 2021. Settling the robust learnability of mixtures of Gaussians. In *STOC '21: 53rd Annual ACM SIGACT Symposium on Theory of Computing, Virtual Event, Italy, June 21–25, 2021*, Samir Khuller and Virginia Vassilevska Williams (Eds.). ACM, 518–531. <https://doi.org/10.1145/3406325.3451084>
- [44] Allen Liu and Ankur Moitra. 2022. Minimax Rates for Robust Community Detection. *arXiv preprint arXiv:2207.11903* (2022).
- [45] Theo McKenzie, Hermish Mehta, and Luca Trevisan. 2020. A New Algorithm for the Robust Semi-random Independent Set Problem. In *Proceedings of the 2020 ACM-SIAM Symposium on Discrete Algorithms, Shuchi Chawla (Ed.)*. 738–746. <https://doi.org/10.1137/1.9781611975994.45>
- [46] Ankur Moitra, William Perry, and Alexander S. Wein. 2016. How robust are reconstruction thresholds for community detection?. In *STOC'16—Proceedings of the 48th Annual ACM SIGACT Symposium on Theory of Computing*. ACM, New York, 828–841. <https://doi.org/10.1145/2897518.2897573>
- [47] Andrea Montanari and Subhabrata Sen. 2016. Semidefinite programs on sparse random graphs and their application to community detection. In *STOC'16—Proceedings of the 48th Annual ACM SIGACT Symposium on Theory of Computing*. ACM, New York, 814–827. <https://doi.org/10.1145/2897518.2897548>
- [48] Prasad Raghavendra and Ning Tan. 2012. Approximating CSPs with global cardinality constraints using SDP hierarchies. In *Proceedings of the Twenty-Third Annual ACM-SIAM Symposium on Discrete Algorithms*. ACM, New York, 373–384.
- [49] Prasad Raghavendra and Morris Yau. 2020. List decodable learning via sum of squares. In *Proceedings of the Fourteenth Annual ACM-SIAM Symposium on Discrete Algorithms*. SIAM, 161–180.
- [50] Prasad Raghavendra and Morris Yau. 2020. List Decodable Subspace Recovery. In *Conference on Learning Theory, COLT 2020, 9–12 July 2020, Virtual Event [Graz, Austria] (Proceedings of Machine Learning Research, Vol. 125)*, Jacob D. Abernethy and Shivani Agarwal (Eds.). PMLR, 3206–3226. <http://proceedings.mlr.press/v125/raghavendra20a.html>
- [51] Tselil Schramm and Alexander S Wein. 2020. Computational barriers to estimation from low-degree polynomials. *arXiv preprint arXiv:2008.02269* (2020).
- [52] Jacob Steinhardt. 2017. Does robustness imply tractability? A lower bound for planted clique in the semi-random model. *arXiv preprint arXiv:1704.05120* (2017).
- [53] Alexander S Wein. 2020. Optimal Low-Degree Hardness of Maximum Independent Set. *arXiv preprint arXiv:2010.06563* (2020).
- [54] David Zuckerman. 2007. Linear degree extractors and the inapproximability of max clique and chromatic number. *Theory Comput.* 3 (2007), 103–128. <https://doi.org/10.4086/toc.2007.v003a006>

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